

ECE 657D: Research Project

Retrieval-Augmented Generation for Question Answering

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Q1: Research Question

This project aims to investigate how the number of retrieved documents (top-k), the length limit for context, and deduplication affect the answer accuracy and hallucination in Retrieval-Augmented Generation (RAG) question answering systems.

Q2: Prior Work and Limitations

[Lewis et al. \[2020\]](#) introduced RAG, which combines a neural retriever with a generator can improve knowledge-intensive question answering by using external documents.

[Karpukhin et al. \[2020\]](#) proposed Dense Passage Retrieval, which improves open-domain QA retrieval performance.

[Mao et al. \[2021\]](#) proposed Generation-Augmented Retrieval (GAR), which integrates retrieval and generation to improve the performance of open-domain question answering.

However, recent RAG surveys [Gao et al. \[2023\]](#) show that the retrieval quality and context selection are critical bottlenecks in real-world RAG systems. Retrieving too many passages can introduce irrelevant information and increase computational cost.

Q3: Novelty of Approach

The previous works on RAG [Lewis et al. \[2020\]](#) focuses mostly on the architecture or large-scale settings while having less discussion on the retrieval configuration.

This project attempts to take a closer look at the hyperparameters for Top-k selection, context length control, and passage deduplication in a small-scale QA-focused RAG system.

Q4: Impact and Motivation

This project is motivated by recent findings [Levy et al. \[2025\]](#) showing that increasing the number of retrieved documents in RAG systems does not necessarily improve performance, suggesting that the effectiveness of RAG depends not only on retrieval itself but also on how retrieved context is selected. Thus, this project focuses on analyzing the impact of Top-k selection, context length control, and passage deduplication on answer accuracy and hallucination in QA-oriented RAG systems.

Q5: Risks and Assumptions

Risks: (1) The quality of the knowledge base and the retrieval embeddings may influence the performance of the RAG system. The answer can be incorrect if the retrieved content does not match the question. (2) there is a limited context window in the language model because the relevant evidence may not be included in the input due to the number of included passages.

Assumptions: (1) The knowledge base contains relevant passages for the QA tasks, and the retrieval model can identify useful evidence. (2) The selected Top-k passages have sufficient information for answer generation within the model's context window.

Q6: Resources

I will use a subset of SQuAD as the dataset, the quantized Llama-3-8B model as the generator, and bge-small-en-v1.5 as embeddings. Experiments will be conducted on Kaggle Notebooks or Google Colab free tier (T4 GPU). Each experiment is estimated to take 1–2 hours depending on dataset size and Top-k configurations.

Q7: Weekly Plan

Week	Plan
1 (Mar 3–7)	[e.g., construct the knowledge base (100 questions). Implement the closed-book baseline QA system.]
2 (Mar 10–14)	[e.g., Implement the simple RAG pipeline, including dense retrieval, passage chunking.]
3 (Mar 17–21)	[e.g., Conduct Top-k experiments and implement context length control.]
4 (Mar 24–28)	[e.g., Implement passage deduplication. Compare performance of baseline, simple RAG, and optimized RAG.]
5 (Mar 31–Apr 2)	[e.g., Analyze results and complete the final report.]

Q8: Milestones and Success Criteria

Milestones: (1) A functional closed-book QA baseline system, (2) A functional simple RAG pipeline with dense retrieval, (3) Implementation of Top-k variation, context length control, and deduplication, (4) Use QA metrics such as Exact Match (EM) and F1 score of closed-book baseline, simple RAG, and optimized RAG.

Success Criteria: (1) The RAG system outperforms the closed-book baseline in QA accuracy and the accuracy of RAG system reaches 80%. (2) Optimized Top-k and context length control demonstrate measurable improvements over the naive RAG setup.

References

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